symmetry principles would have all been discovered by studying its properties. But bear in mind that whereas the equivalence principle gives us some understanding of why gravity exists, and the gauge symmetries give us some sense of why the nongravitational forces exist, in the context of string theory these symmetries are consequences; although their importance is in no way diminished, they are part of the end product of a much larger theoretical structure.

This discussion brings the following question into sharp relief: Is string theory itself an inevitable consequence of some broader principle—possibly but not necessarily a symmetry principle—in much the same way that the equivalence principle inexorably leads to general relativity or that gauge symmetries lead to the nongravitational forces? As of this writing, no one has any insight into the answer to this question. To appreciate its importance, we need only imagine Einstein trying to formulate general relativity without having had the happy thought he experienced in the Bern patent office in 1907 that led him to the principle of equivalence. It would not have been impossible to formulate general relativity without first having this key insight, but it certainly would have been extremely difficult. The equivalence principle provides a succinct, systematic, and powerful organizational framework for analyzing the gravitational force. The description of general relativity we gave in Chapter 3, for example, relied centrally on the equivalence principle, and its role in the full mathematical formalism of the theory is even more crucial.

Currently, string theorists are in a position analogous to an Einstein bereft of the equivalence principle. Since Veneziano’s insightful guess in 1968, the theory has been pieced together, discovery by discovery, revolution by revolution. But a central organizing principle that embraces these discoveries and all other features of the theory within one overarching and systematic framework—a framework that makes the existence of each individual ingredient absolutely inevitable—is still missing. The discovery of this principle would mark a pivotal moment in the development of string theory, as it would likely expose the theory’s inner workings with unforeseen clarity There is, of course, no guarantee that such a fundamental principle exists, but the evolution of physics during the last hundred years encourages string theorists to have high hopes that it does. As we look to the next stage in the